Wheat competition: quality checks and IGE estimation

MOSAIC - IRD experiment 2026

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1 Scope and decisions

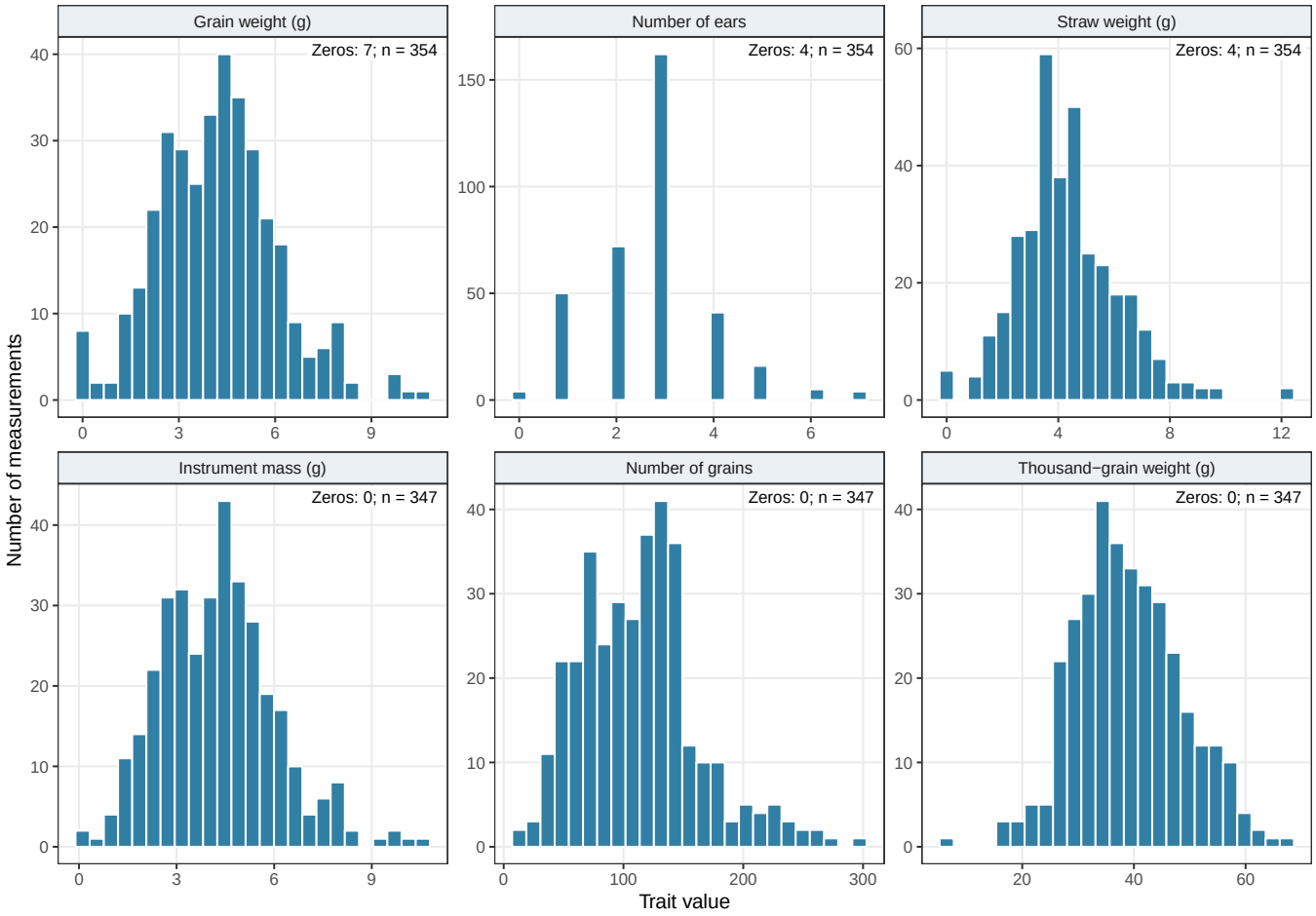
All recorded observations are retained, including zeros, resown plants and harvest comments. Missing responses are not imputed. `resown=1` means that `r` was recorded in the reseeding column; its original `1` denotes later germination, not reseeding. Individual mono plants cannot be matched across source files: their comments/resown flags apply at pot–genotype level.

Preliminary design and pot-type tests use mono/mixt pot means, avoiding treating two plants sharing a pot as independent. IGE models use individual responses and fit within-pot correlation. Instrument mass is omitted from IGE models. Step 3 predictions remain restricted to traits selecting M3.

2 Preliminary quality checks

2.1 Distributions and zero counts

```
distribution<-long %>% group_by(trait) %>% summarise(n=sum(!is.na(value)),
 zeros=sum(value==0,na.rm=TRUE),missing=sum(is.na(value)),.groups='drop') %>%
 mutate(annotation=paste0('Zeros: ',zeros,'; n = ',n))
ggplot(filter(long,!is.na(value)),aes(value))+
 geom_histogram(bins=25,fill='#327FA5',colour='white')+facets()+
 geom_label(data=distribution,aes(x=Inf,y=Inf,label=annotation),
 inherit.aes=FALSE,hjust=1.04,vjust=1.1,size=3,linewidth=0)+
 labs(x='Trait value',y='Number of measurements')
```




```
knitr::kable(select(distribution,trait,n,zeros,missing),row.names=FALSE)
```

trait	n	zeros	missing
poids_grains_g	354	7	0
nb_epis	354	4	0
poids_pailles_g	354	4	0
mass_totale_mesuree_g	347	0	0
nb_grains	347	0	0
PMG	347	0	0

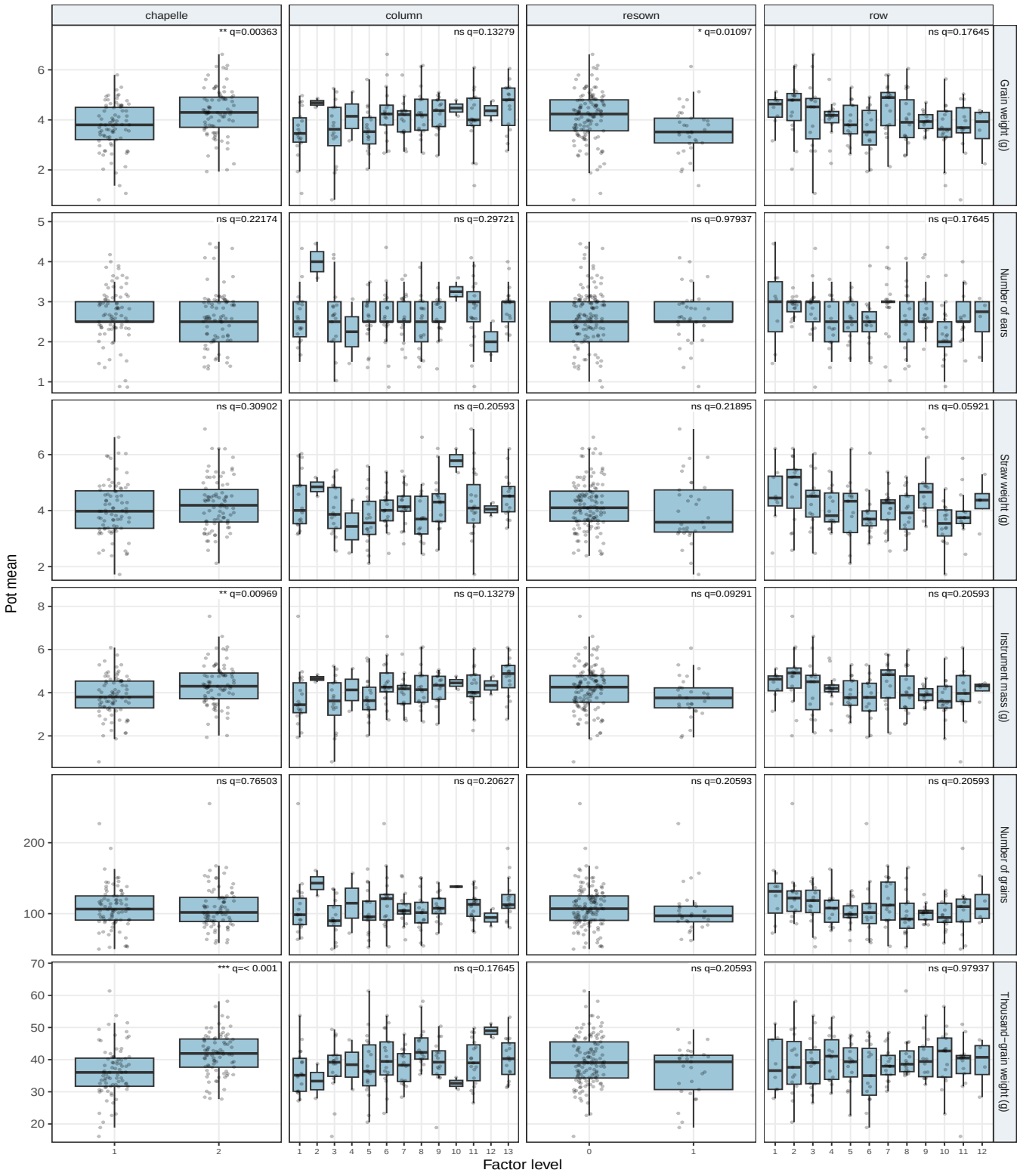
There are 354 manual grain/straw observations and 347 metrology observations. Zero counts are absolute counts, not missing values. The metrology shortfall is absent rows rather than NA values in existing rows. Extreme values are retained. Residual distributions, not raw distributions alone, are relevant to the Gaussian model assumptions.

2.2 Design effects

The following comparisons are marginal Kruskal–Wallis tests. Means based on one available plant receive the same weight as means based on two. A resown pot means at least one genotype is recorded as resown. These limitations are addressed more directly in the subsequent plant-level mixed models.

```
design_factors<-c('chapelle','row','column','resown')
design_data<-bind_rows(lapply(design_factors,function(f){
 pot_means %>% filter(pot_type!='solo',is.finite(value)) %>%
 mutate(factor=f,group=as.character(.data[[f]]))
}))
design_tests<-design_data %>% group_by(trait,factor) %>%
```

```
 summarise(n_pots=n(),p=if(n_distinct(group)>1 && n_distinct(value)>1)
 kruskal.test(value~as.factor(group))$p.value else NA_real_,.groups='drop') %>%
 mutate(q=p.adjust(p,'BH'),annotation=paste0(stars(q),' q=',pfmt(q)))
design_data$group<-factor(design_data$group,
 levels=as.character(sort(unique(as.numeric(design_data$group)))))
ggplot(design_data,aes(group,value))+geom_boxplot(fill='#9DC6D8',outlier.shape=NA)+
 geom_point(position=position_jitter(width=.15,seed=2026),alpha=.25,size=.6)+
 facet_grid(trait~factor,scales='free',labeller=labeller(trait=labels))+
 geom_label(data=design_tests,aes(x=Inf,y=Inf,label=annotation),
 inherit.aes=FALSE,hjust=1.02,vjust=1.05,size=2.3,linewidth=0)+
 scale_y_continuous(expand=expansion(mult=c(.04,.2)))+
 labs(x='Factor level',y='Pot mean')+
 theme(axis.text.x=element_text(size=6),strip.text=element_text(size=8))
```




```
knitr::kable(design_tests %>% select(trait,factor,n_pots,p,q),digits=4)
```

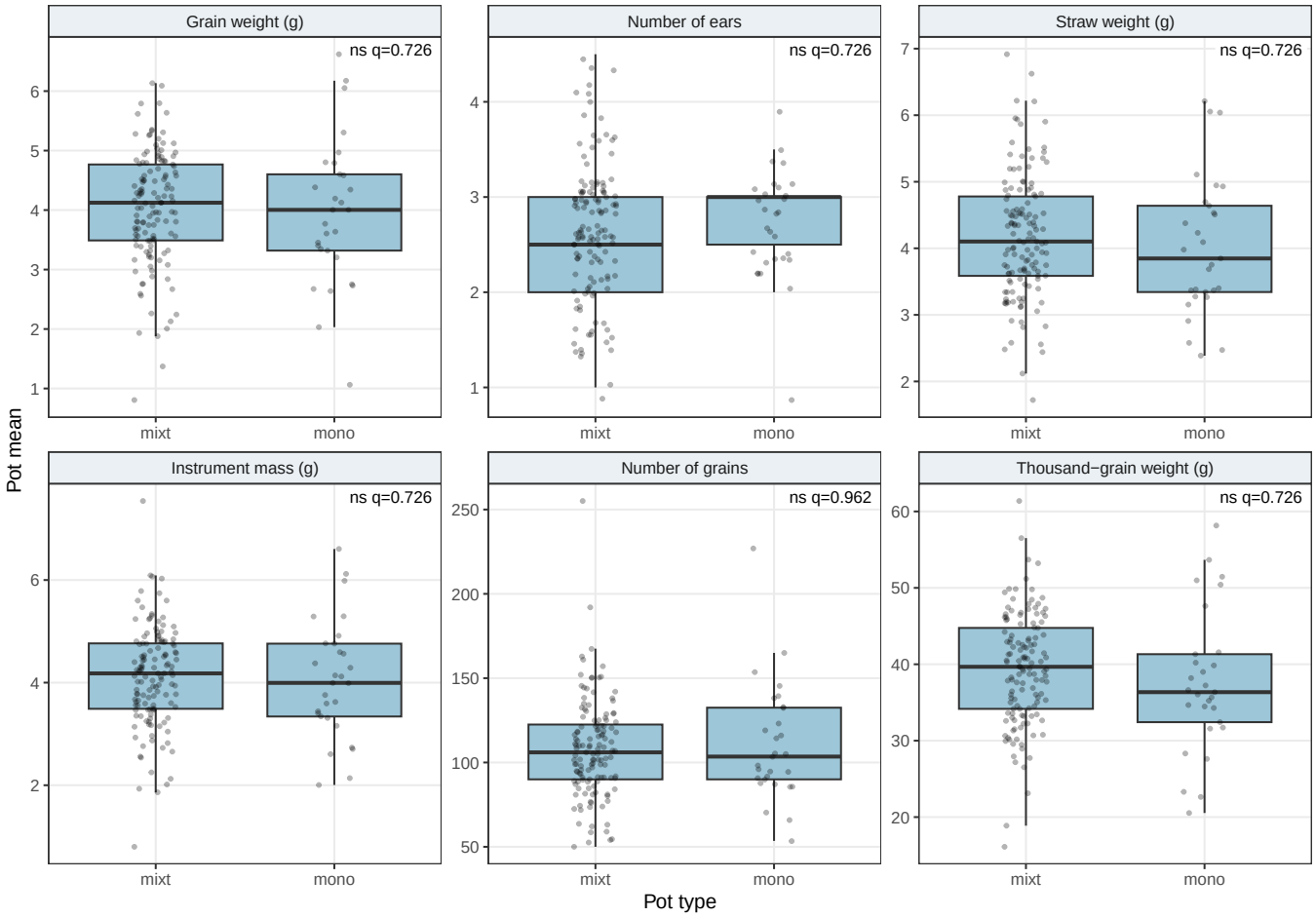
trait	factor	n_pots	p	q
poids_grains_g	chapelle	162	0.0003	0.0036
poids_grains_g	column	162	0.0442	0.1328
poids_grains_g	resown	162	0.0018	0.0110
poids_grains_g	row	162	0.0765	0.1764

trait	factor	n_pots	p	q
nb_epis	chapelle	162	0.1755	0.2217
nb_epis	column	162	0.2477	0.2972
nb_epis	resown	162	0.9504	0.9794
nb_epis	row	162	0.0719	0.1764
poids_pailles_g	chapelle	162	0.2704	0.3090
poids_pailles_g	column	162	0.1292	0.2059
poids_pailles_g	resown	162	0.1642	0.2189
poids_pailles_g	row	162	0.0123	0.0592
mass_totale_mesuree_g	chapelle	162	0.0012	0.0097
mass_totale_mesuree_g	column	162	0.0443	0.1328
mass_totale_mesuree_g	resown	162	0.0232	0.0929
mass_totale_mesuree_g	row	162	0.1078	0.2059
nb_grains	chapelle	162	0.7013	0.7650
nb_grains	column	162	0.1461	0.2063
nb_grains	resown	162	0.1300	0.2059
nb_grains	row	162	0.1373	0.2059
PMG	chapelle	162	0.0000	0.0001
PMG	column	162	0.0809	0.1764
PMG	resown	162	0.1214	0.2059
PMG	row	162	0.9794	0.9794

There are 4 associations with BH q < 0.05 across the 24 tests. These are unadjusted for genotype and other design terms, not causal effects. Ordinal row/column positions include offset end pots and are not measured physical distances.

2.3 Mono versus mixt

```
type_data<-pot_means %>% filter(pot_type!='solo',is.finite(value))
type_tests<-type_data %>% group_by(trait) %>% summarise(
 p=wilcox.test(value~pot_type,exact=FALSE)$p.value,.groups='drop') %>%
 mutate(q=p.adjust(p,'BH'),annotation=paste0(stars(q),' q=',pfmt(q)))
ggplot(type_data,aes(pot_type,value))+geom_boxplot(fill='#9DC6D8',outlier.shape=NA)+
 geom_point(position=position_jitter(width=.12,seed=2026),alpha=.3,size=.8)+facets()+
 geom_label(data=type_tests,aes(x=Inf,y=Inf,label=annotation),
 inherit.aes=FALSE,hjust=1.05,vjust=1.1,size=3,linewidth=0)+
 labs(x='Pot type',y='Pot mean')
```




```
knitr::kable(type_tests %>% select(trait,p,q),digits=4)
```

trait	p	q
poids_grains_g	0.4941	0.7256
nb_epis	0.4147	0.7256
poids_pailles_g	0.3681	0.7256
mass_totale_mesuree_g	0.6047	0.7256
nb_grains	0.9617	0.9617
PMG	0.2005	0.7256

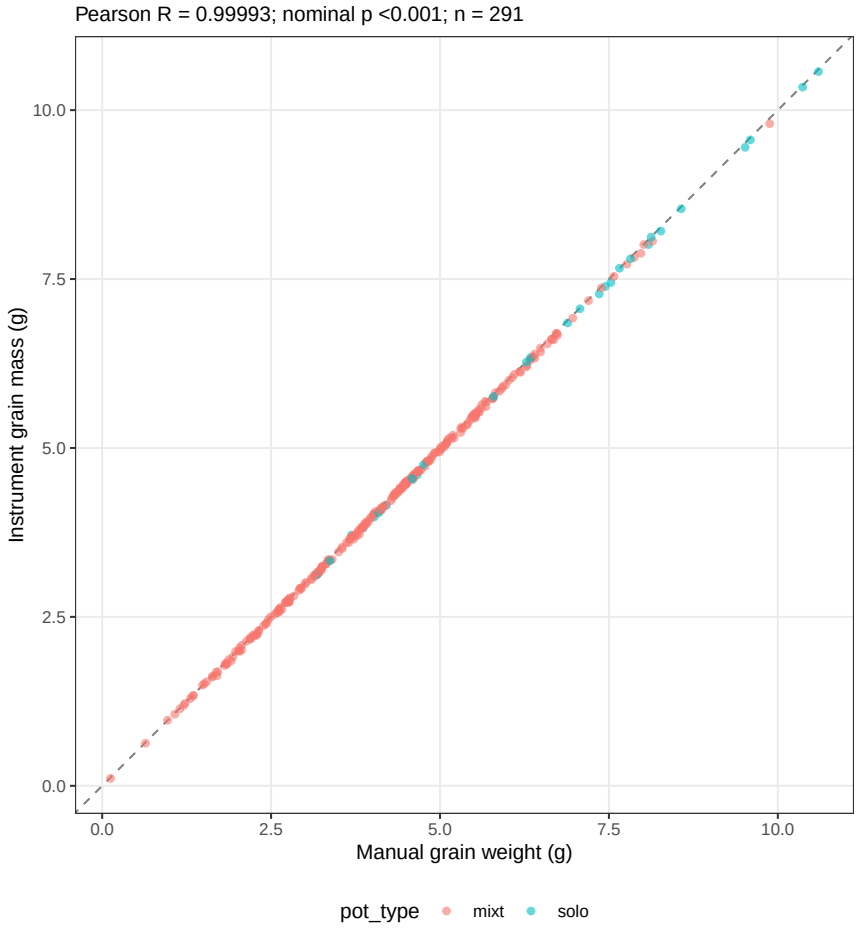
0 traits differ at BH q < 0.05. This marginal comparison differs from the genetically adjusted pot-type fixed-effect test in step 2.

2.4 Grain-weight method comparison

Only solo/mixt plants can be matched unambiguously across source files.

```
keys<-c('pot_id','pot_type','genotype_id','neighbour_id')
a<-grain %>% filter(pot_type!='mono') %>% select(all_of(keys),poids_grains_g)
b<-metro %>% filter(pot_type!='mono') %>% select(all_of(keys),mass_totale_mesuree_g)
stopifnot(!anyDuplicated(a[keys]),!anyDuplicated(b[keys]))
method_pairs<-inner_join(a,b,by=keys) %>%
 filter(is.finite(poids_grains_g),is.finite(mass_totale_mesuree_g))
method_cor<-cor.test(method_pairs$poids_grains_g,
 method_pairs$mass_totale_mesuree_g,method='pearson')
ggplot(method_pairs,aes(poids_grains_g,mass_totale_mesuree_g))+
```

```
 geom_abline(slope=1,intercept=0,linetype=2,colour='grey50')+
 geom_point(aes(colour=pot_type),alpha=.6)+coord_equal()+
 labs(x='Manual grain weight (g)',y='Instrument grain mass (g)',
 subtitle=sprintf('Pearson R = %.5f; nominal p %s; n = %d',
 method_cor$estimate,pfmt(method_cor$p.value),nrow(method_pairs)))
```



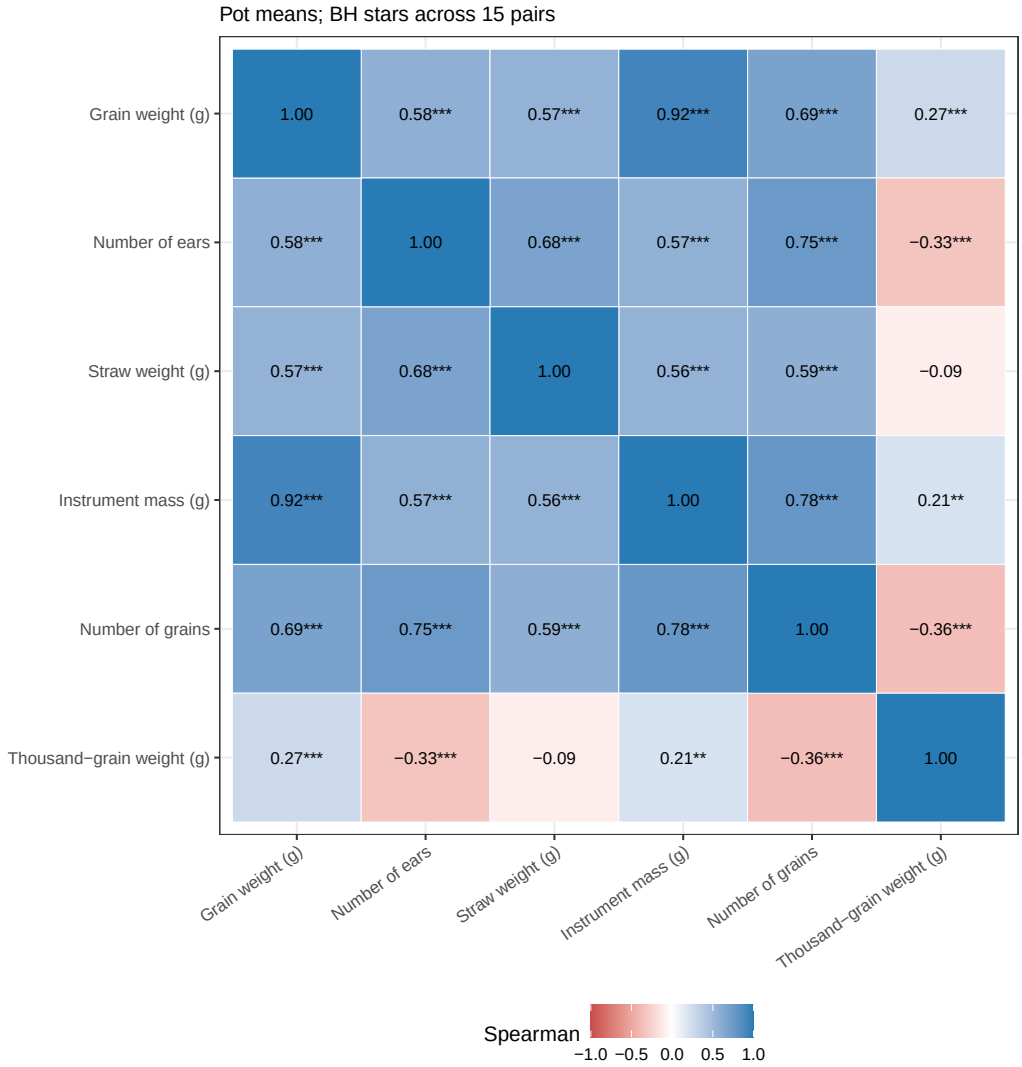
Pearson R = 0.99993 for 291 plants. Correlation is association, not agreement; inspect the equality line too. Nominal inference ignores within-pot dependence. The near duplication motivates omitting instrument mass from genetic models.

2.5 Pairwise trait correlations

Pot means include all pot types without guessing mono plant matches. Spearman correlations are used consistently because several traits are discrete/skewed.

```
wide<-pot_means %>% select(pot_id,trait,value) %>%
 pivot_wider(names_from=trait,values_from=value)
cor_pairs<-bind_rows(lapply(combn(traits6,2,simplify=FALSE),function(pair){
 xy<-wide[,pair];xy<-xy[complete.cases(xy),]
 z<-cor.test(xy[[1]],xy[[2]],method='spearman',exact=FALSE)
 data.frame(first=pair[1],second=pair[2],n=nrow(xy),rho=unname(z$estimate),p=z$p.value)
})) %>% mutate(q=p.adjust(p,'BH'),stars=stars(q))
tiles<-bind_rows(cor_pairs,cor_pairs %>% rename(first=second,second=first),
 data.frame(first=traits6,second=traits6,n=nrow(wide),rho=1,p=NA,q=NA,stars='')) %>%
 mutate(x=factor(first,levels=traits6),y=factor(second,levels=rev(traits6)),
 text=paste0(sprintf('%.2f',rho),ifelse(stars=='ns','',stars)))
ggplot(tiles,aes(x,y,fill=rho))+geom_tile(colour='white')+
```

```
geom_text(aes(label=text),size=3)+coord_fixed()+
scale_fill_gradient2(low='#C84A4A',mid='white',high='#287BB5',limits=c(-1,1))+
scale_x_discrete(labels=labels)+scale_y_discrete(labels=labels)+
theme(axis.text.x=element_text(angle=35,hjust=1))+
labs(x=NULL,y=NULL,fill='Spearman',subtitle='Pot means; BH stars across 15 pairs')
```



14 pairs have BH q < 0.05. Associations pool genotype, spatial and pot-type effects. Incomplete pots use available measurements. They do not identify genetic correlations or causal relationships.

3 IGE estimation

3.1 Equations and terms

For focal genotype i with neighbour j, the shared fixed part is

$$F = \mu + \text{Chapelle}_c + \beta_r \mathbf{1}(\text{resown}) + \beta_m \mathbf{1}(\text{mono}).$$

The final term is omitted when the pot-type testing rule below drops it.

$$\begin{aligned} M1: &\ y_{ip} = F + R_{c,r} + C_{c,k} + D_i + e_{ip}, \\ M2: &\ y_{ip} = F + R_{c,r} + C_{c,k} + D_i + I_j + e_{ip}, \\ M3: &\ y_{ip} = F + R_{c,r} + C_{c,k} + D_i + I_j + S_{ij} + e_{ip}. \end{aligned}$$

$R_{c,r}$ and $C_{c,k}$ are random row and column effects within chapelle; D_i is focal DGE, I_j neighbour IGE and S_{ij} the directed pair effect, distinct from S_{ji}. These have normal distributions with their own variances. In M2/M3 the same genotype's DGE and IGE have covariance

$$(D_g, I_g)' \sim N\left(0, \begin{bmatrix} \sigma^2_D & \sigma_{DI} \\ \sigma_{DI} & \sigma^2_I \end{bmatrix}\right), \quad \sigma_{DI} = \rho_{DI}\sigma_D\sigma_I.$$

Different genotypes are independent: no additive relationship matrix is used. Two residuals sharing a pot have covariance

$$R_p = \sigma^2_e \begin{bmatrix} 1 & \rho_e \\ \rho_e & 1 \end{bmatrix};$$

different pots are independent. Row/column and pair effects are independent of each other and of the DGE/IGE block. Count traits also use Gaussian models.

```
trait_file<-c(poids_grains_g='traits_pesee_grains.csv',nb_epis='traits_pesee_grains.csv',
 poids_pailles_g='traits_pesee_pailles.csv',nb_grains='traits_grain_metrologie.csv',
 PMG='traits_grain_metrologie.csv')
prepare<-function(t){
 d<-rd(trait_file[[t]]);d<-d[d$pot_type!='solo',];d$response<-d[[t]]
 d<-do.call(rbind,lapply(split(d,d$pot_id),function(z){
  miss<-setdiff(1:2,z$plant_id)
  if(length(miss)){
   stopifnot(nrow(z)==1,length(miss)==1)
   extra<-z;extra$plant_id<-miss;extra$response<-NA_real_
   extra$genotype_id<-z$neighbour_id;extra$neighbour_id<-z$genotype_id
   ref<-grain[grain$pot_id==extra$pot_id & grain$genotype_id==extra$genotype_id,]
   extra$resown<-ref$resown[1];z<-rbind(z,extra)
  };z
 }))
 d<-d[order(d$pot_id,d$plant_id),];rownames(d)<-NULL
 g<-sort(unique(c(d$genotype_id,d$neighbour_id)))
 d$DGE<-factor(d$genotype_id,levels=g);d$IGE<-factor(d$neighbour_id,levels=g)
 d$DGExIGE<-factor(paste0('g',d$genotype_id,'_n',d$neighbour_id))
 d$chapelle<-factor(d$chapelle);d$resown<-factor(d$resown)
 d$pot_type<-factor(d$pot_type,levels=c('mixt','mono'))
 d$row_group<-interaction(d$chapelle,d$row,drop=TRUE)
 d$column_group<-interaction(d$chapelle,d$column,drop=TRUE)
 d$pot<-factor(d$pot_id);d$plant<-factor(d$plant_id,levels=1:2)
 stopifnot(!anyDuplicated(d[c('pot','plant')]),all(table(d$pot)==2));d
}
data_list<-setNames(lapply(traits,prepare),traits)
```

NA response slots complete the residual grid; they do not impute phenotypes.

3.2 Pot-type fixed-effect tests and fitting

A conditional asymptotic Wald test (z^2, one degree of freedom) tests the mono coefficient in each M1/M2/M3 fit. This does not compare REML likelihoods with different fixed structures. Raw $p < 0.05$ screens pot type. A trait retains it in all three candidates if any candidate supports it; otherwise all three drop it. Thus each subsequent REML LRT has identical fixed terms. Screening, model selection and variance-parameter uncertainty are not propagated into nominal Wald/LRT p-values.

```
random_forms<-list(M1=~row_group+column_group+DGE,
 M2=~row_group+column_group+str(~DGE+IGE,~corgh(2):id(DGE)),
 M3=~row_group+column_group+str(~DGE+IGE,~corgh(2):id(DGE))+DGExIGE)
fit_model<-function(d,m,with_type){
 fixed<-if(with_type)response~chapelle+resown+pot_type else response~chapelle+resown
 once<-function(gp=list(),rp=list())asreml::asreml(fixed=fixed,
  random=random_forms[[m]],residual=~id(pot):corv(plant),data=d,
```

```
  G.param=gp,R.param=rp,na.action=asreml::na.method(y='include',x='fail'),
  maxit=100,workspace='256mb',trace=FALSE)
 f<-once();if(!isTRUE(f$converge))f<-update(f,maxit=100)
 if(!isTRUE(f$converge))stop('Nonconverged model: ',m)
 if(m!='M1'){
  base<-f
  for(rho in c(-.8,0,.8)){
   gp<-base$G.param
   for(term in names(gp))for(j in seq_along(gp[[term]])){
    b<-gp[[term]][[j]];if(!is.null(b$con))b$con[b$con=='B']<-'P'
    gp[[term]][[j]]<-b
   }
   gp[['DGE+IGE']][['2']]$initial[]<-
    c(rho,.15*var(d$response,na.rm=TRUE),.05*var(d$response,na.rm=TRUE))
   gp[['DGE+IGE']][['2']]$con<-c('U','P','P')
   candidate<-try(once(gp,base$R.param),silent=TRUE)
   valid<-!inherits(candidate,'try-error') && isTRUE(candidate$converge)
   if(valid && candidate$loglik>f$loglik)f<-candidate
  }
 };f
}
```

```
full_models<-list();pot_tests<-list();keep_type<-setNames(logical(length(traits)),traits)
for(t in traits){
 full_models[[t]]<-setNames(lapply(names(random_forms),function(m)
  fit_model(data_list[[t]],m,TRUE)),names(random_forms))
 pot_tests[[t]]<-bind_rows(lapply(names(random_forms),function(m){
  co<-summary(full_models[[t]][[m]],coef=TRUE)$coef.fixed
  ix<-match('pot_type_mono',rownames(co));stopifnot(!is.na(ix))
  estimate<-co[ix,'solution'];se<-co[ix,'std error']
  data.frame(trait=t,model=m,mono_minus_mixt=estimate,SE=se,
   p=pchisq((estimate/se)^2,1,lower.tail=FALSE))
 }))
 keep_type[t]<-any(pot_tests[[t]]$p<.05)
}
```

```
## Online License checked out Fri Oct 23 17:05:15 2026
```

```
pot_tests<-bind_rows(pot_tests)
knitr::kable(pot_tests,digits=4,row.names=FALSE)
```

trait	model	mono_minus_mixt	SE	p
poids_grains_g	M1	0.2157	0.1900	0.2562
poids_grains_g	M2	0.2044	0.1885	0.2781
poids_grains_g	M3	0.2049	0.2257	0.3639
nb_epis	M1	0.1543	0.1418	0.2764
nb_epis	M2	0.1543	0.1418	0.2765
nb_epis	M3	0.1561	0.1481	0.2918
poids_pailles_g	M1	0.0756	0.1582	0.6330
poids_pailles_g	M2	0.0697	0.1579	0.6588
poids_pailles_g	M3	0.0718	0.1689	0.6707
nb_grains	M1	7.0316	4.6215	0.1281
nb_grains	M2	6.9880	4.5930	0.1281
nb_grains	M3	7.0083	4.7350	0.1388
PMG	M1	-0.7070	1.0196	0.4881
PMG	M2	-0.7071	1.0196	0.4880
PMG	M3	-0.7059	1.0369	0.4960

Pot type retained for: **none of the traits**. Dropping it does not prove its effect is zero; the allocation of average mono–mixt differences among genetic terms changes.

3.3 Model selection

M1–M2 jointly tests IGE variance and DGE–IGE covariance, using the asymptotic mixture $0.5\chi^2_1 + 0.5\chi^2_2$. M2–M3 tests one interaction variance using $0.5\chi^2_0 + 0.5\chi^2_1$. With ten genotypes and boundary estimates these approximations warrant caution. Select M3 if M3–M2 has $p < 0.05$; otherwise select M2 if M2–M1 has $p < 0.05$, else M1. BH across ten tests is supplementary.

```
models<-list();lrt<-list();selection<-list()
for(t in traits){
 models[[t]]<-if(keep_type[t])full_models[[t]] else
  setNames(lapply(names(random_forms),function(m)fit_model(data_list[[t]],m,FALSE)),
   names(random_forms))
 compare<-function(a,b,joint){
  delta<-b$loglik-a$loglik
  if(delta< -1e-3)stop('Nested likelihood ordering failed: ',t)
  LR<-max(0,2*delta)
  p<-if(LR==0)1 else if(joint)
   .5*pchisq(LR,1,lower.tail=FALSE)+.5*pchisq(LR,2,lower.tail=FALSE) else
   .5*pchisq(LR,1,lower.tail=FALSE)
  data.frame(LR=LR,p=p)
 }
 a<-compare(models[[t]]$M1,models[[t]]$M2,TRUE)
 b<-compare(models[[t]]$M2,models[[t]]$M3,FALSE)
 best<-if(b$p<.05)'M3' else if(a$p<.05)'M2' else 'M1'
 selection[[t]]<-data.frame(trait=t,pot_type_retained=keep_type[t],selected=best)
 lrt[[t]]<-rbind(cbind(data.frame(trait=t,comparison='M1 to M2'),a),
  cbind(data.frame(trait=t,comparison='M2 to M3'),b))
}
selection<-bind_rows(selection);lrt<-bind_rows(lrt) %>% mutate(q=p.adjust(p,'BH'))
knitr::kable(selection,row.names=FALSE)
```

trait	pot_type_retained	selected
poids_grains_g	FALSE	M3
nb_epis	FALSE	M1
poids_pailles_g	FALSE	M1
nb_grains	FALSE	M1
PMG	FALSE	M1

```
knitr::kable(lrt,digits=5,row.names=FALSE)
```

trait	comparison	LR	p	q
poids_grains_g	M1 to M2	5.80335	0.03546	0.17732
poids_grains_g	M2 to M3	3.27762	0.03512	0.17732
nb_epis	M1 to M2	0.01333	0.95072	0.97897
nb_epis	M2 to M3	0.13759	0.35534	0.51189
poids_pailles_g	M1 to M2	1.92630	0.27343	0.51189
poids_pailles_g	M2 to M3	0.18732	0.33258	0.51189
nb_grains	M1 to M2	1.94435	0.27073	0.51189
nb_grains	M2 to M3	0.13173	0.35832	0.51189
PMG	M1 to M2	0.00261	0.97897	0.97897
PMG	M2 to M3	0.02506	0.43711	0.54639

Selected models: poids_grains_g M3; nb_epis M1; poids_pailles_g M1; nb_grains M1; PMG M1. There are 2 nominally significant and 0 BH-significant tests. These data do not warrant treating interaction selection as robust without sensitivity or bootstrap checks.

3.4 Marginal phenotypic variance shares

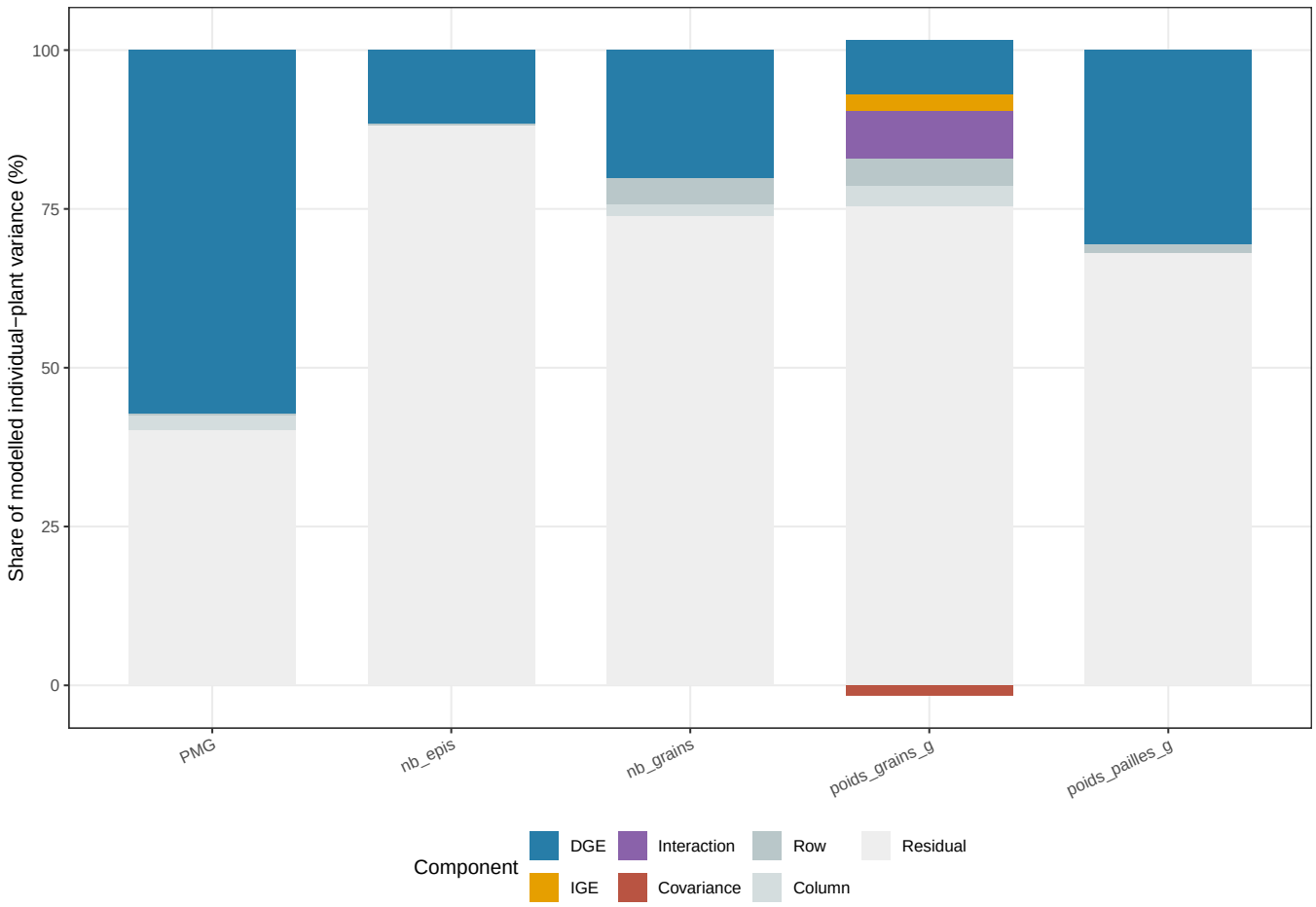
The average conditional individual-plant variance, over the observed mono/mixt composition, is

$$V_P = \sigma^2_D + \sigma^2_I + \sigma^2_S + 2p_{mono}\sigma_{DI} + \sigma^2_R + \sigma^2_C + \sigma^2_e.$$

p_{mono} is the fraction of observed plant responses in mono pots. In mixt pots different genotype effects are independent; in mono pots the same genotype supplies DGE and IGE, adding $2\sigma_{DI}$. Each component's share is $100v_k/V_P$. The covariance share is $100(2p_{mono}\sigma_{DI})/V_P$ and can be negative. Signed shares sum to 100%. Within-pot residual correlation changes between-plant covariance, not individual variance, and is not an additional variance component. Shares condition on fixed effects: they are not fractions of raw sample variance, total breeding-value variance, or causal contributions.

```
getv<-function(vc,n)as.numeric(vc[n,'component'])
genetic_block<-function(f,m){
 vc<-summary(f)$varcomp
 if(m=='M1')return(c(D=getv(vc,'DGE'),I=0,cov=0,rho=NA))
 vd<-getv(vc,'DGE+IGE!corgh(2)_1');vi<-getv(vc,'DGE+IGE!corgh(2)_2')
 ix<-startsWith(rownames(vc),'DGE+IGE!') & endsWith(rownames(vc),'.cor')
 rho<-as.numeric(vc[ix,'component']);stopifnot(length(rho)==1)
 c(D=vd,I=vi,cov=rho*sqrt(vd*vi),rho=rho)
}
variance<-bind_rows(lapply(traits,function(t){
 m<-selection$selected[selection$trait==t];f<-models[[t]][[m]]
 vc<-summary(f)$varcomp;g<-genetic_block(f,m);d<-data_list[[t]]
 pm<-mean(d$genotype_id[!is.na(d$response)]==d$neighbour_id[!is.na(d$response)])
 v<-c(DGE=g[['D']],IGE=g[['I']],Interaction=if(m=='M3')getv(vc,'DGExIGE')else 0,
  Covariance=2*pm*g[['cov']],Row=getv(vc,'row_group'),
  Column=getv(vc,'column_group'),Residual=as.numeric(vc[endsWith(rownames(vc),'!var'),'component']))
 data.frame(trait=t,component=names(v),variance=as.numeric(v),percent=100*v/sum(v))
}))
```

```
variance$component<-factor(variance$component,
 levels=c('DGE','IGE','Interaction','Covariance','Row','Column','Residual'))
ggplot(variance,aes(trait,percent,fill=component))+geom_col(width=.7)+
 scale_fill_manual(values=c('#277DA8','#E69F00','#8A62AA','#B95442',
 '#B9C7C9','#D4DDDE','#EEEEEE'))+
 labs(x=NULL,y='Share of modelled individual-plant variance (%)',fill='Component')+
 theme(axis.text.x=element_text(angle=25,hjust=1))
```




```
knitr::kable(variance %>% filter(component %in% c('DGE','IGE','Interaction','Covariance')) %>%
 select(trait,component,percent),digits=3,row.names=FALSE)
```

trait	component	percent
poids_grains_g	DGE	8.619
poids_grains_g	IGE	2.525
poids_grains_g	Interaction	7.625
poids_grains_g	Covariance	-1.595
nb_epis	DGE	11.592
nb_epis	IGE	0.000
nb_epis	Interaction	0.000
nb_epis	Covariance	0.000
poids_pailles_g	DGE	30.630
poids_pailles_g	IGE	0.000
poids_pailles_g	Interaction	0.000
poids_pailles_g	Covariance	0.000
nb_grains	DGE	20.183
nb_grains	IGE	0.000
nb_grains	Interaction	0.000
nb_grains	Covariance	0.000
PMG	DGE	57.241
PMG	IGE	0.000
PMG	Interaction	0.000
PMG	Covariance	0.000

3.5 Genetic predictions and joint model-based errors

Only M3-selected traits are exported. The DGE baseline is $B_i = \mu + D_i$. The directed interaction contribution is $H_{ij} = I_j + S_{ij} + \beta_m \mathbf{1}(i = j)$ when pot type is retained, and $I_j + S_{ij}$ when it is dropped. The interaction file has a single combined interaction column. Adding the baseline and interaction gives the fitted reference-condition one-neighbour expectation (chapelle 1, non-resown, zero random row/column effects). The baseline is not solo yield: it comes from the competition model with neighbour effects set to their population means.

Joint errors must retain covariances between fixed and genetic terms and between different predictions. For $\theta = (\beta', u')'$, the error variance of $a'\widehat{\theta}$ is $a'C_\theta a$, not the sum of component SEs. With $V = ZGZ' + R$, define

$$C_\beta = (X'V^{-1}X)^{-1}, \quad K = GZ'V^{-1}, \quad B = KX.$$

The joint BLUE/BLUP error covariance is

$$C_\theta = \begin{bmatrix} C_\beta & -C_\beta B' \\ -BC_\beta & G - KZG + BC_\beta B' \end{bmatrix}.$$

We reconstruct this matrix from ASReml variance estimates and the observed-data design, then check coefficients and their standard errors against ASReml. For the joint prediction vector, with contrast matrix L,

$$C_{pred} = LC_\theta L'.$$

The prediction tables report $\sqrt{\mathrm{diag}(C_{pred})}$ and approximate 95% bounds (estimate ± 1.96 SE). The complete joint covariance is exported per trait, with matching `prediction_id` labels. A grid prediction with weights w has model-based error variance $w'C_{pred}w$, including reuse of the same effect across several neighbours.

These errors condition on estimated variance parameters and the selected model. They exclude variance-parameter/model-selection uncertainty and the uncertainty of transferring one-neighbour effects to a grid. Bounds describe latent expectations, not intervals for a new plant's phenotype; future residual variation is not added. The mono fixed contrast, if retained, is experimental pot-type adjustment, not a calibrated response to multiple same-genotype neighbours.

```
joint_errors<-function(f,d,m,with_type){
 d<-d[!is.na(d$response),]
 fixed<-if(with_type)~chapelle+resown+pot_type else ~chapelle+resown
 X<-model.matrix(fixed,d);stopifnot(qr(X)$rank==ncol(X))
 xnames<-colnames(X)
 for(term in c('chapelle','resown','pot_type'))
  xnames<-sub(paste0('^',term),paste0(term,'_'),xnames)
 colnames(X)<-xnames
 zfactor<-function(v,term){
  z<-model.matrix(~v-1);colnames(z)<-paste0(term,'_',levels(v));z
 }
 Zparts<-list(zfactor(d$row_group,'row_group'),
  zfactor(d$column_group,'column_group'),zfactor(d$DGE,'DGE'))
 vc<-summary(f)$varcomp;g<-genetic_block(f,m)
 ng<-nlevels(d$DGE);nr<-nlevels(d$row_group);nc<-nlevels(d$column_group)
 if(m!='M1')Zparts<-c(Zparts,list(zfactor(d$IGE,'IGE')))
 if(m=='M3')Zparts<-c(Zparts,list(zfactor(d$DGExIGE,'DGExIGE')))
 Z<-do.call(cbind,Zparts);G<-matrix(0,ncol(Z),ncol(Z))
 diag(G)<-c(rep(getv(vc,'row_group'),nr),rep(getv(vc,'column_group'),nc),
  rep(g[['D']],ng),if(m!='M1')rep(g[['I']],ng),
  if(m=='M3')rep(getv(vc,'DGExIGE'),nlevels(d$DGExIGE)))
 if(m!='M1'){
  di<-nr+nc+seq_len(ng);ii<-nr+nc+ng+seq_len(ng)
  G[cbind(di,ii)]<-g[['cov']];G[cbind(ii,di)]<-g[['cov']]
 }
 rho<-as.numeric(vc[endsWith(rownames(vc),'!plant!cor'),'component'])
 ve<-as.numeric(vc[endsWith(rownames(vc),'!var'),'component'])
 R<-ve*((1-rho)*diag(nrow(d))+rho*outer(d$pot_id,d$pot_id,'=='))
```

```
 V<-Z%*%G%*%t(Z)+R;Vi<-chol2inv(chol(V))
 Cb<-solve(crossprod(X,Vi%*%X));K<-G%*%t(Z)%*%Vi;B<-K%*%X
 beta<-as.numeric(Cb%*%crossprod(X,Vi%*%d$response))
 u<-as.numeric(K%*%(d$response-X%*%beta))
 Cu<-G-K%*%Z%*%G+B%*%Cb%*%t(B)
 C<-rbind(cbind(Cb,-Cb%*%t(B)),cbind(-B%*%Cb,Cu));C<-(C+t(C))/2
 cn<-c(colnames(X),colnames(Z));dimnames(C)<-list(cn,cn)
 theta<-setNames(c(beta,u),cn)
 cf<-summary(f,coef=TRUE)$coef.fixed;cr<-summary(f,coef=TRUE)$coef.random
 ref<-c(cf[match(colnames(X),rownames(cf)),'solution'],
  cr[match(colnames(Z),rownames(cr)),'solution'])
 ref_se<-c(cf[match(colnames(X),rownames(cf)),'std error'],
  cr[match(colnames(Z),rownames(cr)),grep('^std[ .]error$',colnames(cr),value=TRUE)[1]])
 difference<-max(abs(theta-ref),na.rm=TRUE)
 se_difference<-max(abs(sqrt(pmax(0,diag(C)))-ref_se),na.rm=TRUE)
 stopifnot(difference<.005,se_difference<.005)
 list(theta=theta,C=C,estimate_check=difference,SE_check=se_difference)
}
prediction_dge<-prediction_interaction<-error_checks<-list()
for(t in selection$trait[selection$selected=='M3']){
 d<-data_list[[t]];f<-models[[t]]$M3;with_type<-keep_type[t]
 j<-joint_errors(f,d,'M3',with_type);g<-levels(d$DGE)
 pairs<-expand.grid(genotype=g,neighbour=g,stringsAsFactors=FALSE)
 id_d<-paste0('DGE:',t,':',g)
 id_h<-paste0('interaction:',t,':',pairs$genotype,':',pairs$neighbour)
 L<-matrix(0,length(g)+nrow(pairs),length(j$theta),
  dimnames=list(c(id_d,id_h),names(j$theta)))
 for(i in seq_along(g))L[i,c('(Intercept)',paste0('DGE_',g[i]))]<-1
 for(i in seq_len(nrow(pairs))){
  ix<-length(g)+i
  L[ix,c(paste0('IGE_',pairs$neighbour[i]),
   paste0('DGExIGE_g',pairs$genotype[i],'_n',pairs$neighbour[i]))]<-1
  if(with_type && pairs$genotype[i]==pairs$neighbour[i])L[ix,'pot_type_mono']<-1
 }
 values<-as.numeric(L%*%j$theta);CP<-L%*%j$C%*%t(L)
 stopifnot(min(eigen(CP,symmetric=TRUE,only.values=TRUE)$values)> -1e-6)
 se<-sqrt(pmax(0,diag(CP)));nd<-length(g);ih<-nd+seq_len(nrow(pairs))
 prediction_dge[[t]]<-data.frame(trait=t,genotype=g,prediction_id=id_d,
  intercept=j$theta[['(Intercept)']],DGE_BLUP=j$theta[paste0('DGE_',g)],
  prediction=values[seq_len(nd)],prediction_formula='intercept + DGE',
  prediction_SE=se[seq_len(nd)],
  lower95=values[seq_len(nd)]-1.96*se[seq_len(nd)],
  upper95=values[seq_len(nd)]+1.96*se[seq_len(nd)],
  reference_chapelle=levels(d$chapelle)[1],reference_resown=levels(d$resown)[1],
  reference_pot_type=if(with_type)'mixt' else 'not fitted',row.names=NULL)
 prediction_interaction[[t]]<-data.frame(trait=t,pairs,prediction_id=id_h,
  interaction=values[ih],
  prediction_formula=ifelse(with_type & pairs$genotype==pairs$neighbour,
    'IGE + DGE:IGE + mono fixed effect','IGE + DGE:IGE'),
  prediction_SE=se[ih],
  lower95=values[ih]-1.96*se[ih],upper95=values[ih]+1.96*se[ih],
  pot_type_adjustment_included=with_type & pairs$genotype==pairs$neighbour)
 write_excel(data.frame(prediction_id=rownames(CP),CP,check.names=FALSE),
  file.path(ige_dir,paste0('prediction_error_covariance_',t,'.csv')))
 error_checks[[t]]<-data.frame(trait=t,max_estimate_difference=j$estimate_check,
  max_SE_difference=j$SE_check)
}
```

```
DGE<-if(length(prediction_dge))bind_rows(prediction_dge)else
 data.frame(trait=character(),genotype=character(),prediction=numeric(),prediction_SE=numeric())
interactions<-if(length(prediction_interaction))bind_rows(prediction_interaction)else
 data.frame(trait=character(),genotype=character(),neighbour=character(),
 interaction=numeric(),prediction_SE=numeric())
write_excel(DGE,file.path(ige_dir,'03_DGE_genotype_predictions.csv'))
write_excel(interactions,file.path(ige_dir,'03_all_genotype_interactions.csv'))
knitr::kable(bind_rows(error_checks),digits=7,row.names=FALSE)
```

trait	max_estimate_difference	max_SE_difference
poids_grains_g	0.0001587	4.88e-05

```
knitr::kable(DGE %>% select(trait,genotype,prediction,prediction_SE),digits=4,row.names=FALSE)
```

trait	genotype	prediction	prediction_SE
poids_grains_g	35	4.6817	0.2849
poids_grains_g	117	3.4150	0.2894
poids_grains_g	184	4.0120	0.2860
poids_grains_g	198	3.7892	0.2878
poids_grains_g	233	3.7965	0.2886
poids_grains_g	315	4.3105	0.2851
poids_grains_g	372	4.1590	0.2836
poids_grains_g	402	3.8490	0.2837
poids_grains_g	447	3.4101	0.2842
poids_grains_g	476	3.9335	0.2860

The two main prediction files contain 10 baselines and 100 directed interactions. The covariance file supplies the error covariance for their concatenated vector. Baselines have the intercept added; interactions do not. Same-genotype and different-genotype predictions are together, and only same-genotype predictions include a mono contrast if that fixed term was retained. No individual genotype CSVs are generated.

3.5.1 Solo BLUEs versus competition DGE predictions

The `prediction_formula` column states which fitted terms enter each exported prediction. The baseline is `intercept + DGE`; interaction predictions contain neighbour IGE plus the directed pair effect, with a mono fixed adjustment only if that term was retained. Prediction SEs use the joint errors for exactly those terms.

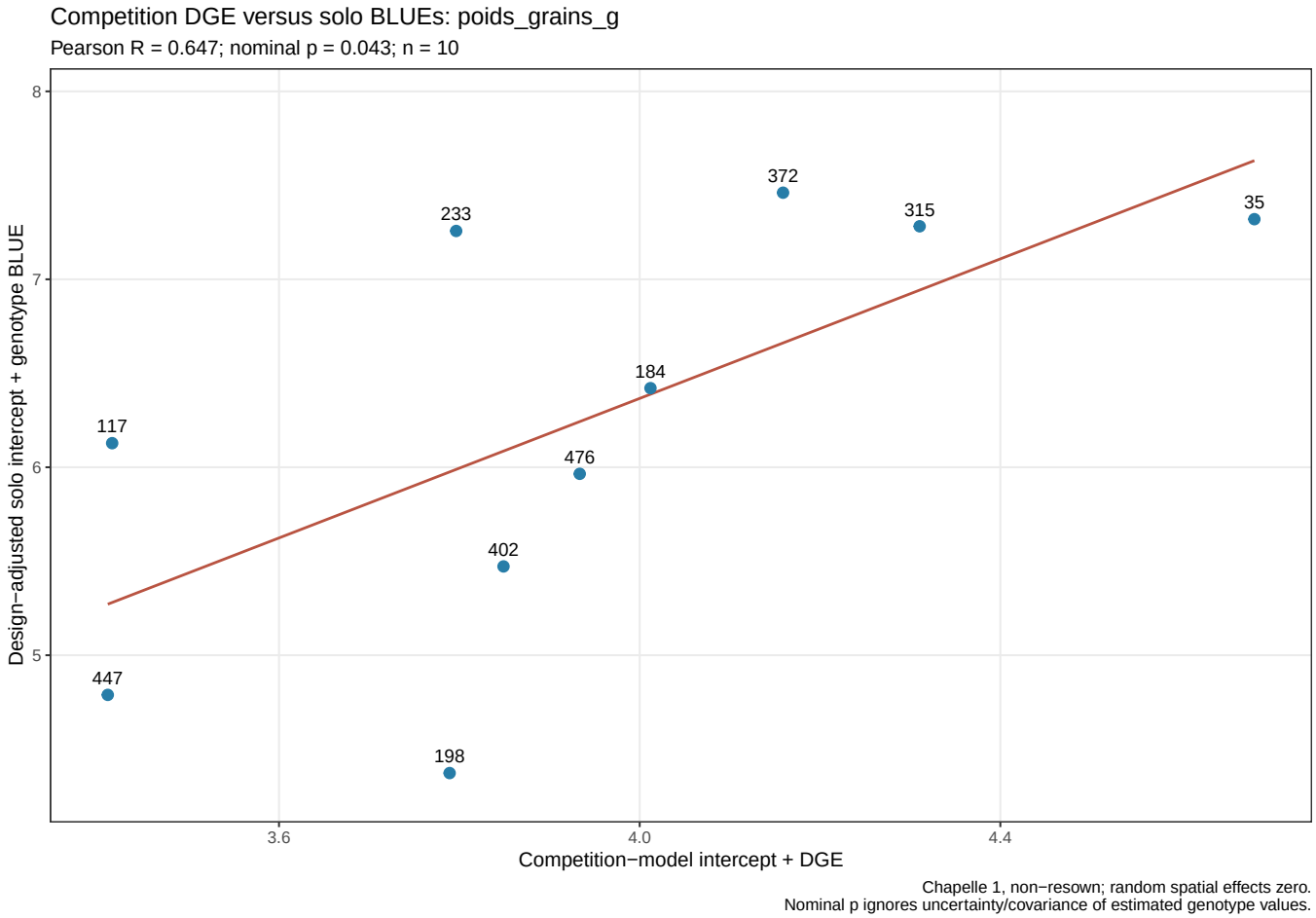
For comparison with solo performance, genotype is fitted as a **fixed effect** in a separate ASReml model using all available solo observations:

$$y_{ip} = \mu_s + \text{Chapelle}_c + \beta_{r,s}\mathbf{1}(\text{resown}) + G_i + R_{c,r} + C_{c,k} + e_{ip}.$$

The solo expectation is the intercept plus genotype BLUE. Chapelle and resown are fixed, row/column within chapelle are random, and residuals are independent. Pot type cannot be fitted within a solo-only dataset and no within-pot residual correlation is estimable. Both comparisons refer to chapelle 1 and non-resown plants, with random spatial effects set to zero.

```
# Solo BLUEs use fixed genotype and all applicable design corrections.
# Pot type is constant for solo plants; no residual pair exists within a pot.
solo_comparisons<-solo_correlations<-list()
for(t in unique(DGE$trait)){
 solo<-rd(trait_file[[t]])
 solo<-solo[solo$pot_type=='solo' & is.finite(solo[[t]]),]
 solo$response<-solo[[t]]
 solo$genotype<-factor(solo$genotype_id,
   levels=sort(unique(solo$genotype_id)))
```

```
 solo$chapelle<-factor(solo$chapelle)
 solo$resown<-factor(solo$resown)
 solo$row_group<-interaction(solo$chapelle,solo$row,drop=TRUE)
 solo$column_group<-interaction(solo$chapelle,solo$column,drop=TRUE)
 solo_fit<-asreml::asreml(fixed=response~chapelle+resown+genotype,
   random=~row_group+column_group,residual=~idv(units),data=solo,
   na.action=asreml::na.method(y='fail',x='fail'),
   maxit=100,workspace='256mb',trace=FALSE)
 if(!isTRUE(solo_fit$converge))solo_fit<-update(solo_fit,maxit=100)
 stopifnot(isTRUE(solo_fit$converge),!anyDuplicated(solo$pot_id))
 co<-summary(solo_fit,coef=TRUE)$coef.fixed
 g<-levels(solo$genotype)
 ix<-match(paste0('genotype_',g),rownames(co))
 intercept_ix<-match('(Intercept)',rownames(co))
 ref_ix<-match(c(paste0('chapelle_',levels(solo$chapelle)[1]),
   paste0('resown_',levels(solo$resown)[1])),rownames(co))
 stopifnot(!anyNA(c(ix,intercept_ix,ref_ix)),
   all(co[ref_ix,'solution']==0))
 expected<-data.frame(genotype=g,
   solo_BLUE=co[intercept_ix,'solution']+co[ix,'solution'])
 baseline<-DGE[DGE$trait==t,c('genotype','prediction')]
 baseline$genotype<-as.character(baseline$genotype)
 matched<-merge(baseline,expected,by='genotype')
 stopifnot(nrow(matched)==length(g),!anyDuplicated(matched$genotype))
 test<-cor.test(matched$prediction,matched$solo_BLUE,method='pearson')
 solo_correlations[[t]]<-data.frame(trait=t,n_genotypes=nrow(matched),
   Pearson_R=unname(test$estimate),p_nominal=test$p.value)
 solo_comparisons[[t]]<-matched
 p<-ggplot(matched,aes(prediction,solo_BLUE))+
   geom_smooth(method='lm',formula=y~x,se=FALSE,
     colour='#B95442',linewidth=.7)+
   geom_point(size=2.5,colour='#277DA8')+
   geom_text(aes(label=genotype),vjust=-.8,size=3.5)+
   scale_y_continuous(expand=expansion(mult=c(.08,.15)))+
   labs(title=paste('Competition DGE versus solo BLUEs:',t),
     subtitle=sprintf('Pearson R = %.3f; nominal p = %s; n = %d',
       test$estimate,pfmt(test$p.value),nrow(matched)),
     x='Competition-model intercept + DGE',
     y='Design-adjusted solo intercept + genotype BLUE',
     caption='Chapelle 1, non-resown; random spatial effects zero.\nNominal p ignores uncertainty/covariance
 print(p)
}
```




```
solo_correlations<-dplyr::bind_rows(solo_correlations)
knitr::kable(solo_correlations,digits=4,row.names=FALSE)
```

trait	n_genotypes	Pearson_R	p_nominal
poids_grains_g	10	0.6475	0.043

poids_grains_g: Pearson R = 0.647, nominal p = 0.043 (10 genotypes).

A positive correlation indicates that genotypes with larger DGE expectations also tend to have greater solo performance. This is not a test of equality: adding an intercept does not affect Pearson correlation, and high correlation can coexist with different absolute yield levels. The p-value tests zero correlation, not whether the correlation equals one.

Perfect correlation is not expected for several reasons:

- Solo performance and competition DGE are different quantities. The DGE baseline is estimated in two-plant pots with neighbour effects centred at zero; it is not an extrapolated absence-of-neighbour prediction.
- The two analyses use different observations and density/competition contexts. Each solo genotype has only three plants, making its adjusted mean noisy.
- Solo genotype effects are BLUEs without genetic shrinkage. Competition DGEs are BLUPs with shrinkage, informed by the joint DGE/IGE covariance and pair interaction model. Uniform shrinkage alone would preserve correlation; here information and adjustment can differ among genotypes.
- Design and variance parameters are estimated separately. Sampling error, unequal adjustments and genuine context-dependent genotype performance can all contribute. This plot cannot separate those explanations.

The correlation is descriptive: its nominal Pearson p-value ignores uncertainty and covariance in the estimated genotype values. It does not by itself prove that the underlying solo and competition genetic effects differ or agree.

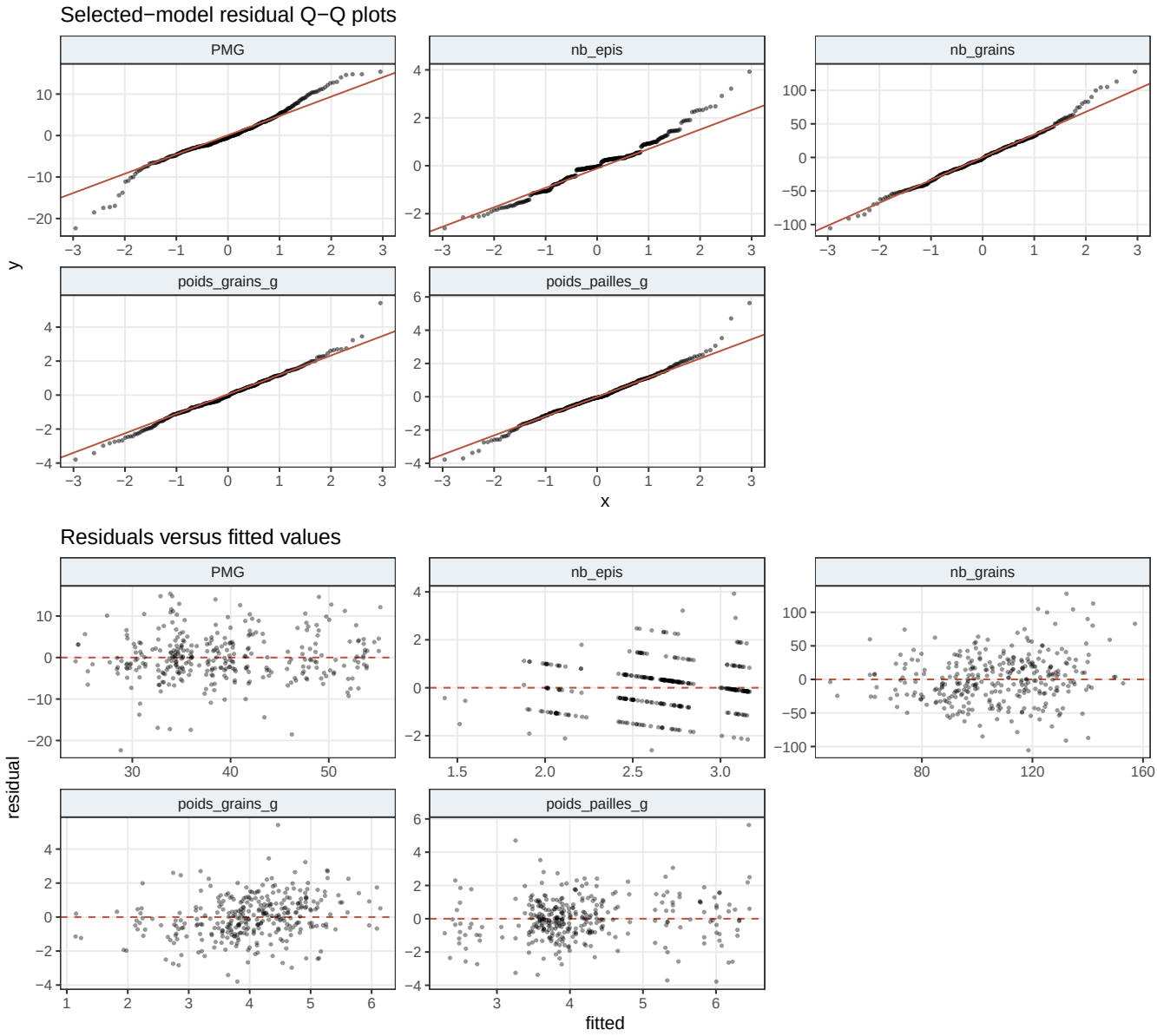
3.6 Numerical and residual diagnostics

```
diagnostics<-bind_rows(lapply(traits,function(t){
 bind_rows(lapply(names(random_forms),function(m){
  f<-models[[t]][[m]];g<-genetic_block(f,m)
  data.frame(trait=t,model=m,converged=f$converge,loglik=f$loglik,
   DGE_IGE_correlation=g[['rho']])
 }))
}))
knitr::kable(diagnostics,digits=4,row.names=FALSE)
```

trait	model	converged	loglik	DGE_IGE_correlation
poids_grains_g	M1	TRUE	-291.4903	NA
poids_grains_g	M2	TRUE	-288.5886	-0.7304
poids_grains_g	M3	TRUE	-286.9498	-0.9552
nb_epis	M1	TRUE	-178.2254	NA
nb_epis	M2	TRUE	-178.2187	-0.9991
nb_epis	M3	TRUE	-178.1499	-0.9988
poids_pailles_g	M1	TRUE	-256.4926	NA
poids_pailles_g	M2	TRUE	-255.5295	-0.7500
poids_pailles_g	M3	TRUE	-255.4358	-0.7892
nb_grains	M1	TRUE	-1309.8743	NA
nb_grains	M2	TRUE	-1308.9021	-0.4096
nb_grains	M3	TRUE	-1308.8362	-0.4210
PMG	M1	TRUE	-733.8150	NA
PMG	M2	TRUE	-733.8137	0.9589
PMG	M3	TRUE	-733.8011	0.9589

```
residual_data<-bind_rows(lapply(traits,function(t){
 m<-selection$selected[selection$trait==t];f<-models[[t]][[m]];d<-data_list[[t]]
 ok<-!is.na(d$response)
 data.frame(trait=t,residual=as.numeric(f$residuals)[ok],
 fitted=as.numeric(f$linear.predictors)[ok])
}))
```

```
p1<-ggplot(residual_data,aes(sample=residual))+stat_qq(size=.6,alpha=.5)+
 stat_qq_line(colour='#B95442')+facet_wrap(~trait,scales='free',ncol=3)+
 labs(title='Selected-model residual Q-Q plots')
p2<-ggplot(residual_data,aes(fitted,residual))+geom_hline(yintercept=0,
 linetype=2,colour='#B95442')+geom_point(size=.6,alpha=.4)+
 facet_wrap(~trait,scales='free',ncol=3)+labs(title='Residuals versus fitted values')
patchwork::wrap_plots(p1,p2,ncol=1)
```



All 15 final models report convergence: **TRUE**. Multiple starting correlations were tried for M2/M3. Convergence does not validate the model assumptions. Q-Q plots of raw conditional residuals are descriptive: shrinkage and pot correlation mean they are not independent, identically distributed residual samples. Tail departures, especially for PMG and counts, and changing residual spread require investigation. Correlations near +/-1 make genetic components hard to separate. Bootstrap LRTs and influence analyses would strengthen inference.

4 Deliverables and reproducibility

This report holds the code, main tests, figures and interpretation. The subfolders retain only figures and the genetic prediction/covariance tables. Intermediate inputs, model RDS objects and the old without-pot-type comparison are not needed to use the final predictions. Rerendering this report fits the models anew from processed data and requires ASReml-R licensing, the listed R packages, Pandoc and LaTeX.

Genetic effects are total genotype-level effects, not additive-only effects. The exported normal-approximation errors describe within-model precision; they do not justify assuming additive or rank-independent interactions in a many-neighbour grid. Resowing is adjusted statistically; it is not a filter.

```
## R version 4.5.1 (2025-06-13 ucrt)
## Platform: x86_64-w64-mingw32/x64
```

```
## Running under: Windows 11 x64 (build 26300)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=C                          LC_CTYPE=English_United States.utf8
## [3] LC_MONETARY=C                         LC_NUMERIC=C
## [5] LC_TIME=C
##
## time zone: Europe/Paris
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
## [1] tidyr_1.3.1   dplyr_1.1.4   ggplot2_4.0.1
##
## loaded via a namespace (and not attached):
##  [1] Matrix_1.7-3       gtable_0.3.6       compiler_4.5.1     tidyselect_1.2.1
##  [5] splines_4.5.1      scales_1.4.0       yaml_2.3.11        fastmap_1.2.0
##  [9] lattice_0.22-7     R6_2.6.1           patchwork_1.3.2    labeling_0.4.3
## [13] generics_0.1.4     knitr_1.50         tibble_3.3.0       asreml_4.2.0.392
## [17] pillar_1.11.1      RColorBrewer_1.1-3 rlang_1.1.6        xfun_0.54
## [21] S7_0.2.1           cli_3.6.5          withr_3.0.2        magrittr_2.0.4
## [25] mgcv_1.9-3         digest_0.6.39      grid_4.5.1         lifecycle_1.0.5
## [29] nlme_3.1-168       vctrs_0.6.5        evaluate_1.0.5     glue_1.8.0
## [33] data.table_1.17.8  farver_2.1.2       rmarkdown_2.30     purrr_1.2.0
## [37] tools_4.5.1        pkgconfig_2.0.3    htmltools_0.5.9
```

4.1 Documentation

ASReml REML likelihood tests: https://asreml.kb.vsni.co.uk/knowledge-base/lrt-asreml/

ASReml prediction functions: https://asreml.kb.vsni.co.uk/knowledge-base/predict-asreml/